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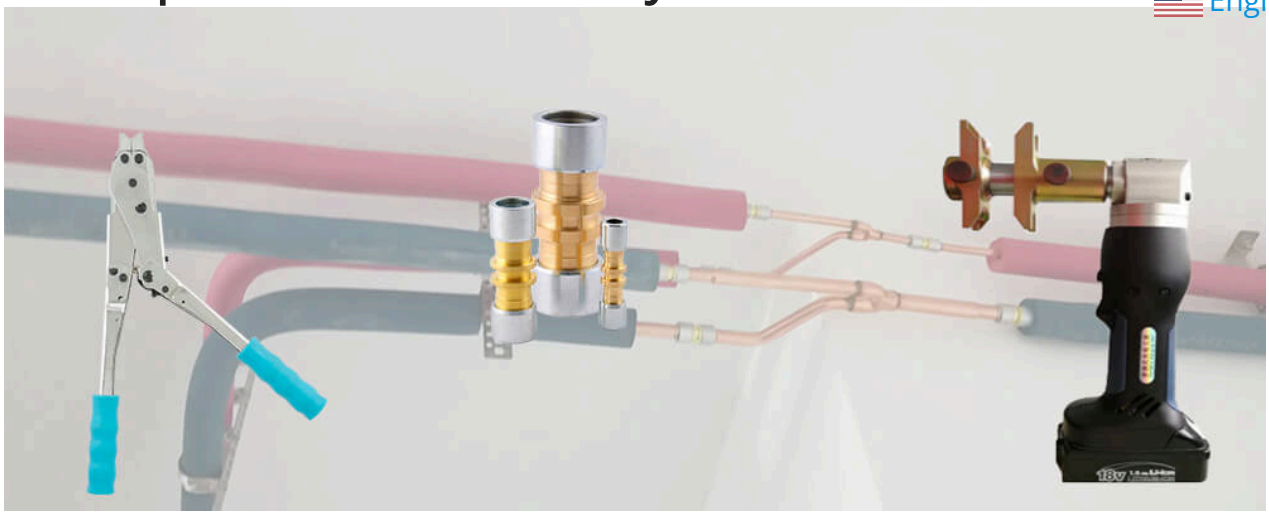
A Complete Guide for VRF System Installation

English ▾



Pic VRF unit uses Miracle Lokring Tool and Fittings

The VRF (Variable Refrigerant Flow) system, also known as VRV (Variable Refrigerant Volume), is increasingly becoming the most preferred choice for modern HVAC solutions in buildings

This advanced VRF HVAC technology offers room-by-room heating and cooling, providing a efficient climate control system.

Nowadays, VRF systems are considered a panacea for buildings, highly favored by technicians, design engineers, and facility managers for their precision, efficiency, and flexibility.

The VRF system installation process is different from split C, so from building operators to installers and technicians, ensuring proper VRF installation following the manufacturer's VRF system installation manuals and maintaining installation quality is a must; otherwise, problems arise.

Here are full guidelines, best practices, and safety tips for your VRF installation project success.

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English ▾

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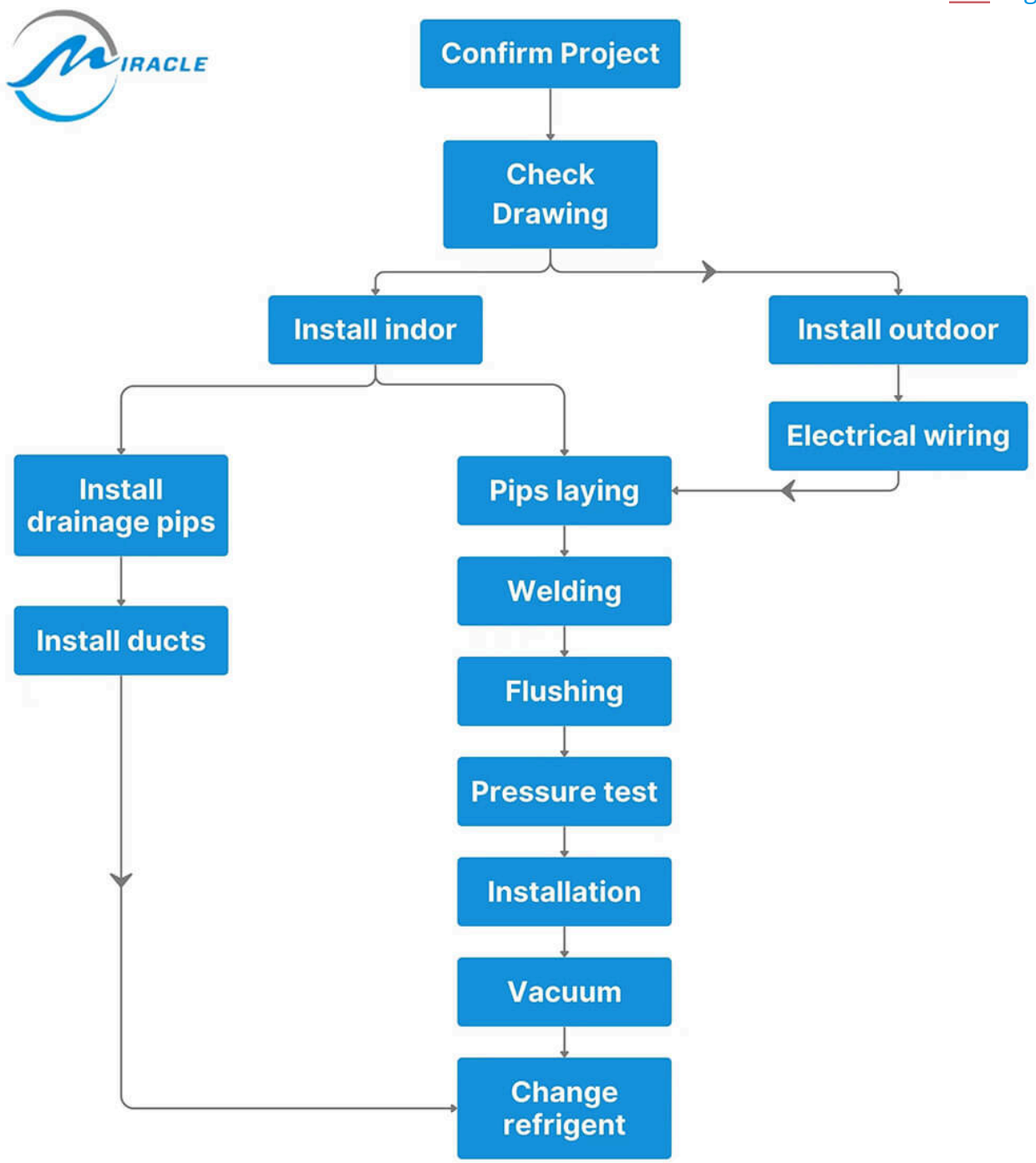
English ▾



Preparing for VRF System Installation

While planning your VRF installation, considering the following critical factors is a must to ensure a successful and efficient installation:

Pipe Size Selection:

English ▾

Selecting the appropriate VRF pipe size is crucial for optimal VRF system performance. This ensures efficient refrigerant flow, minimizes pressure drops, and enhances overall system efficiency.

To determine the size of refrigerant pipes, pay close attention to the VRF project building's specific requirements, total piping length, and system capacity.

2. Piping Route Finalization:

Finalizing the VRF piping routes is another vital step because unplanned piping route selection brings unnecessary bends and long runs, which can affect the system's performance.

Pipe should not be placed in areas like elevator shafts, enclosed public stairways, or emergency exit door places.

To ensure you comply with the global VRF pipework route, you can comply with ASHRAE 15 and 34, which address the safe installation of the HVAC system because this standard addresses safety measures like refrigerant leakage, which might affect gas displacement ar personal safety.

3. Indoor and Outdoor Unit Location:

The VRF indoor unit should be in a location where air distribution is even, and maintenance access is easy.

The outdoor VRF unit should be in well-ventilated areas to ensure efficient heat exchange and avoid obstructions that could hamper airflow.

At the same time, ensure that the line length is within VRF manufacturer specifications.

4. Choosing the Right Air Conditioning Condensate Drain Pump

English ▾

After determining the cooling capacity of the indoor unit and its installation location, you need to consider the route of the air conditioner condensate drain line.

You need to choose the right air conditioning drain pump based on the cooling capacity of the indoor unit and then develop a reasonable drain line route based on the type of air conditioning drain pump.

Types of air conditioner drain pumps include VRF condensate pumps, low-profile condensa pumps, mini split condensate pumps, wall-mounted condensate pumps, and tank condensate pumps.

The air conditioner brand can be divided into LG condensate pump, GREE condensate pum Mitsubishi condensate pump, Daikin condensate pump, Midea condensate pump, York condensate pump, etc.

Different types of condensate pumps are used in different environments, check the blog condensate pumps a complete guide, or contact us now to choose the right condensate pump for your air conditioner.

Check more types of air conditioner condensate drain pumps.

4. Other Consideration:

You also need to choose the ventilation system and the right type of VRF system according to the building's heating and cooling requirements.

If the building utilizes a building management system BMS, ensure its compatibility with the F system.

English ▾

Essential Tools for VRF System Installation

There are many types of HVAC tools for VRF unit installation, repair, and maintenance.

For an ideal HVAC technician or business owner, it's a good idea to choose more portable tools instead of manual and traditional tools.

There is a comprehensive set of tools you need to use to make your VRF system installatior smooth, efficient, and safe, including measurement tools, pipe and tube handling tools, electric tools, safety equipment tools, vacuum and charging equipment, maintenance, and diagnostic tools, etc.

Before installing the VRF system, it's always best practice to wear safety equipment like safety shoes, glasses, racing glasses, sleeves, safety helmets, gloves, safety belts, etc, to protect against injury from falls.

You also need refrigerant gas, refrigerant cylinder, refrigerant charging meter, manifold gauge, charging hose, vacuum pump, vacuum gauge, HVAC flaring tool, safety valve, pressure regulator, level gauge, anti-vibration rubber, multimeter, phase check, etc.

To download the full HVAC list categorized into basic technician tools, no welding connection tools, and specialized HVAC safety tools, click here and save it. [HVAC tools pdf linking]

VRF Indoor Unit Installation

English ▾

The first is to select the installation position of the VRF indoor unit. According to the room configuration, select the best optimum direction of supply airflow and use the plumb bob to determine the location.

As the wiring and piping will be on the side wall surface, the maintenance access door shou be placed in the proper space.

The more space you will provide, the more efficiently you can work with safety.

To prepare the hanging structure for an indoor unit, use the unit dimension, fixing points, and pump bulb to confirm that the indoor unit body is handled with hanging bolts correctly in place.

After confirming the unit's position and fixing the hanging bolt before tightening the nuts to fix the hanging bolts, please use a level gauge to check that the indoor unit is perfectly horizontal.

A perfectly leveled unit can reduce running noise. In the ductwork process, insert a canvas duct between the indoor unit and duct to provide full insulation to the inlet flange and outlet duct to prevent condensation.

Related Reading: Top VRF Manufacturers with the Best Sales in the USA

VRF Outdoor Unit Installation:

English ▾

If you need to install the VRF outdoor unit at a high place, the lifting procedure should include using robust suspension ropes that are capable of bearing the weight of the VRF outdoor unit.

Use protective pads in this process to protect the outdoor unit from being scratched. After placing the VRF outdoor unit on the surface:

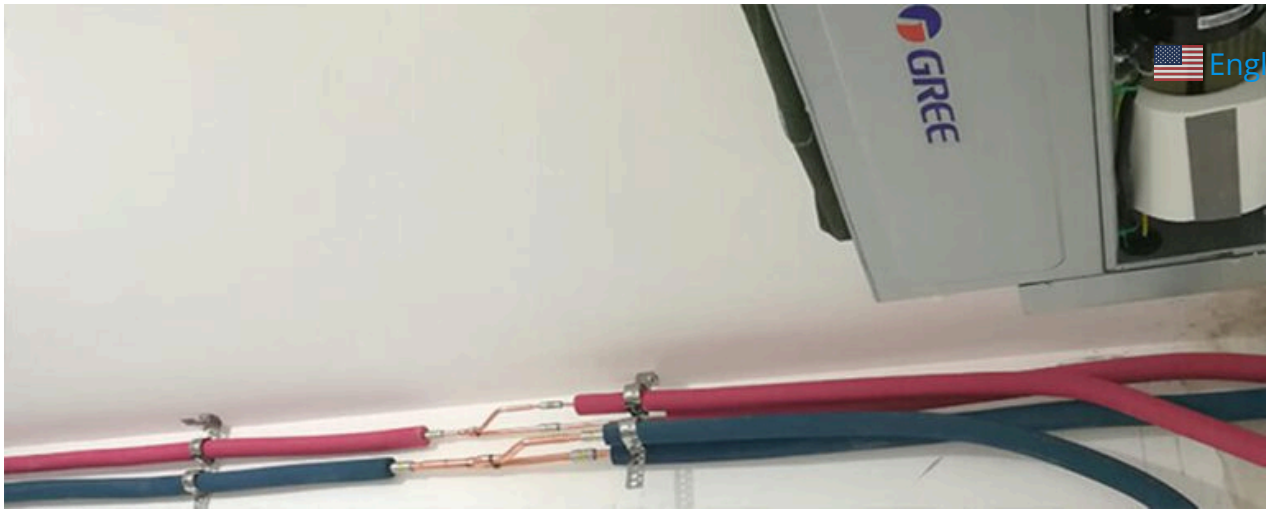
1. Use a concrete base or angle bracket as the unit's foundation and stabilize it as much as possible so it does not move to the side.
2. Provide sufficient vibration proofing, like an anti-vibration pad between the base and, th is large enough to cover the entire width of each unit's legs.
3. Make sure all bottom corners of the unit are firmly seated and the outdoor unit is level.

Otherwise, the unit's leg may bend, preventing it from standing in balance. Keep all of the outdoor units on the same level in the ground.

Don't choose a place to install VRF outdoor units where there are potentially explosive atmospheres, electromagnetic waves, the risk of fire, carbon fiber, ignitable dust, and corrosive gas.

Related Reading: Top 10VRF Manufacturers in China

VRF Piping Installation



English ▾

The performance of the VRF system is closely related to the pipeline. First, you need to choose the right pipeline material. Currently, copper and aluminum pipelines are popular in the market.

Secondly, check whether there are impurities in the connection of the VRF pipeline. For example, if welding is used, a large amount of welding slag will appear inside the pipeline, thus blocking the VRF indoor and outdoor units.

We recommend using the HVAC press tool and fittings. It is the most reliable VRF pipeline connection solution.

Finally, check the insulation pipe of the VRF. Ensure the thickness and density of the VRF pi insulation, to effectively isolate the generation of condensed water and prevent the pipelin from rusting.

Please learn more about the precautions for VRF pipe here.

Flushing VRF System

English ▾

When the piping connection with the indoor is complete, and before setting up the final connection, the essential task is Flushing.

It removes all dust, particles, and moisture that can cause compressor malfunction.

Simultaneously flush the liquid and gas sides with 0.5 MPa nitrogen, block the opening of tl pipe with your hand when it's in high pressure, and then remove it.

This is a repeated process until there is no more dust or moisture in the pipe, and then the opening of the pipe is sealed.

The sequence of this Flushing process for the VRF installation starts from the indoor unit ar moves forward to the outdoor unit, sealing all openings one by one when the flushing is complete.

VRF System Gas Tightness Test

Gas tightness testing is a crucial process to ensure there are no leaks in refrigerant lines containing gases.

English ▾

This test also uses nitrogen to pressurize the gas pimples to a specific pressure and then monitors for any pressure drops that would indicate a leak.

This test needs to take place when the piping system is fully completed with both VRF indoor and VRF outdoor units and the gas stop valves are closed.

VRF System Vacuum Drying

A leak-free refrigeration system is key to successful VRF system installation, which can be achieved by pressure testing by vacuuming the refrigerant.

If any other water, moisture, or contaminant is present in the refrigerant pipe, this vacuum drying technique should be followed.

The vacuum pump needs to be capable of -0.1 Mpa pressurizing or less than that.

VRF System Charging Refrigerant

English ▾

After these two tests, it's time to charge the refrigerant while maintaining the proper amount.

Use the diameter and the liquid pipe length to measure the proper amount of refrigerant gas that is needed.

The process to add more refrigerant is to calculate the proper amount of refrigerant, turn c all of the liquid and gas stop valves, and charge the refrigerant. Stop the refrigerant chargir when the amount reaches the gas amount you require.

Use the same refrigerant that your compressor and other tools are compatible with. Remember to use the proper safety tools and equipment that we mentioned in the safety tools section.

VRF Drain Piping & Electrical Wiring

The drain pipe should be separate from other water pipes, shorter in length, and not joining with other pipes that are dirty.

English ▾

According to the rules, the main pipe must have a larger slope than 1%. After the drain piping, electrical power supply wiring, and installation of the controller are the next steps.

If the indoor unit is connected to one outdoor system, you need to use a single power supply. Otherwise, if the outdoor unit is running but the indoor unit is shut off, the evaporator of the indoor unit would continuously flow the refrigerant.

Still, the fan is stopped, which would turn to freeze. This causes a shortage of refrigerant gas to run the unit properly and finally damages the compressor.

Related Product: Air-conditioner-drain-pump

Commissioning and Testing VRF System

Before commissioning the VRF system, a test run is very important to confirm the system is working properly.

There are several tests that you need to run and test one by one. Some of them are airtight tests, water tightness and flow tests, power supply tests, insulation resistance tests, refrigerant charge verification, etc.

Before starting the test run, check for loose cables and connectors.

Check the voltage of each system component after the power supply is turned on. Follow these steps to complete the commissioning process of the VRF system:

English ▾

- First, connect all monitoring controllers and tools, such as the BC port, remote controllers, time clocks, and centralized controllers.
- Then, turn on each indoor unit one by one, checking both cooling and heating modes to confirm the system is working at accurate temperature conditions. Inspect the full system operation for at least one hour and keep a detailed record of all data.
- Finally, you can complete the commissioning process after verifying the successful installation and operation of the VRF system.

VRF/VRV System Installation Cost:

VRV/VRF installation costs consist of three parts. Please check VRF System Cost.

- VRFIndoor unit installation cost:
- VRFOutdoor Installation Cost:
- VRF Piping Installation Cost:

The cost of indoor and outdoor units varies greatly depending on the brand and power.

Please learn more about the VRF unit brands we have compiled.

If you need to purchase VRF indoor and outdoor units. Please send it to the Miracle team and we will provide you with competitive prices.

Another important VRV installation cost is the installation of the pipes.

There are two parts of the cost, VRF copper pipe material cost and installation cost.

Miracle recommends that you contact us directly to purchase copper pipe insulation.

In addition, you should also use our braze-free HVAC fittings.

English ▾

These two combinations can effectively reduce the installation and use costs of your VRF system. Contact us now.

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